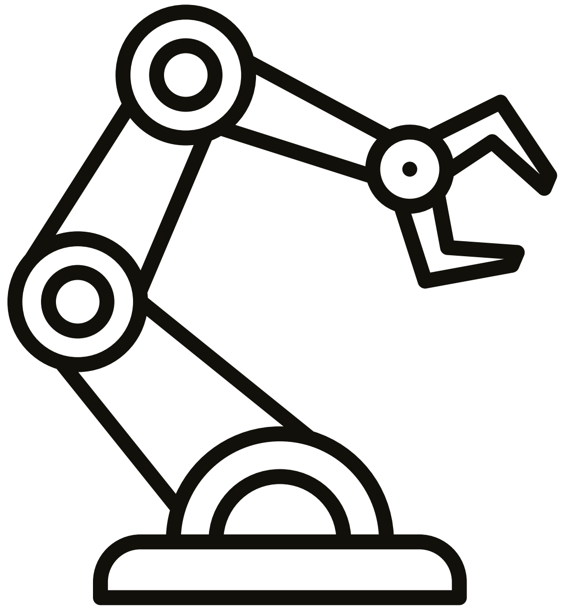
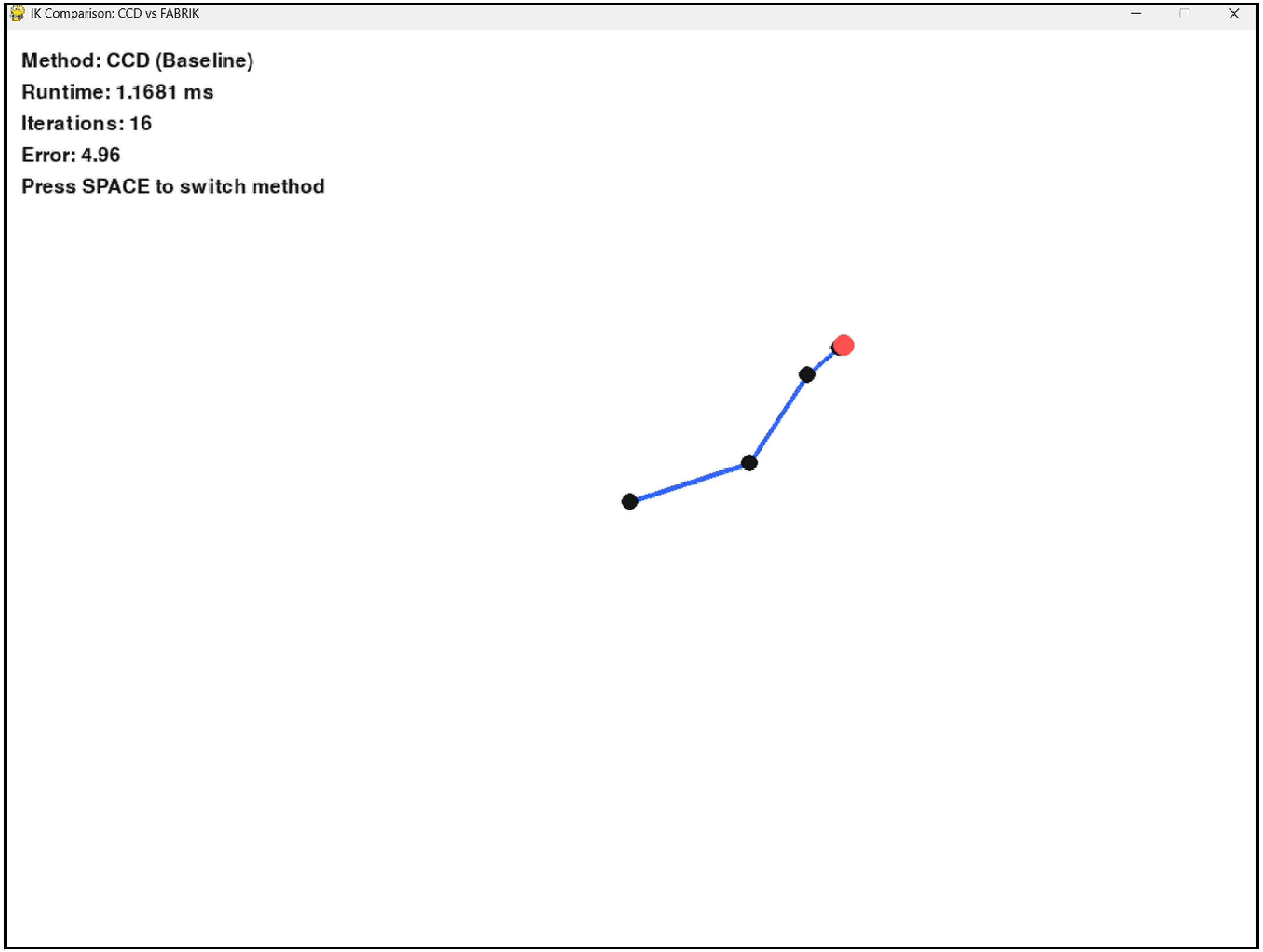
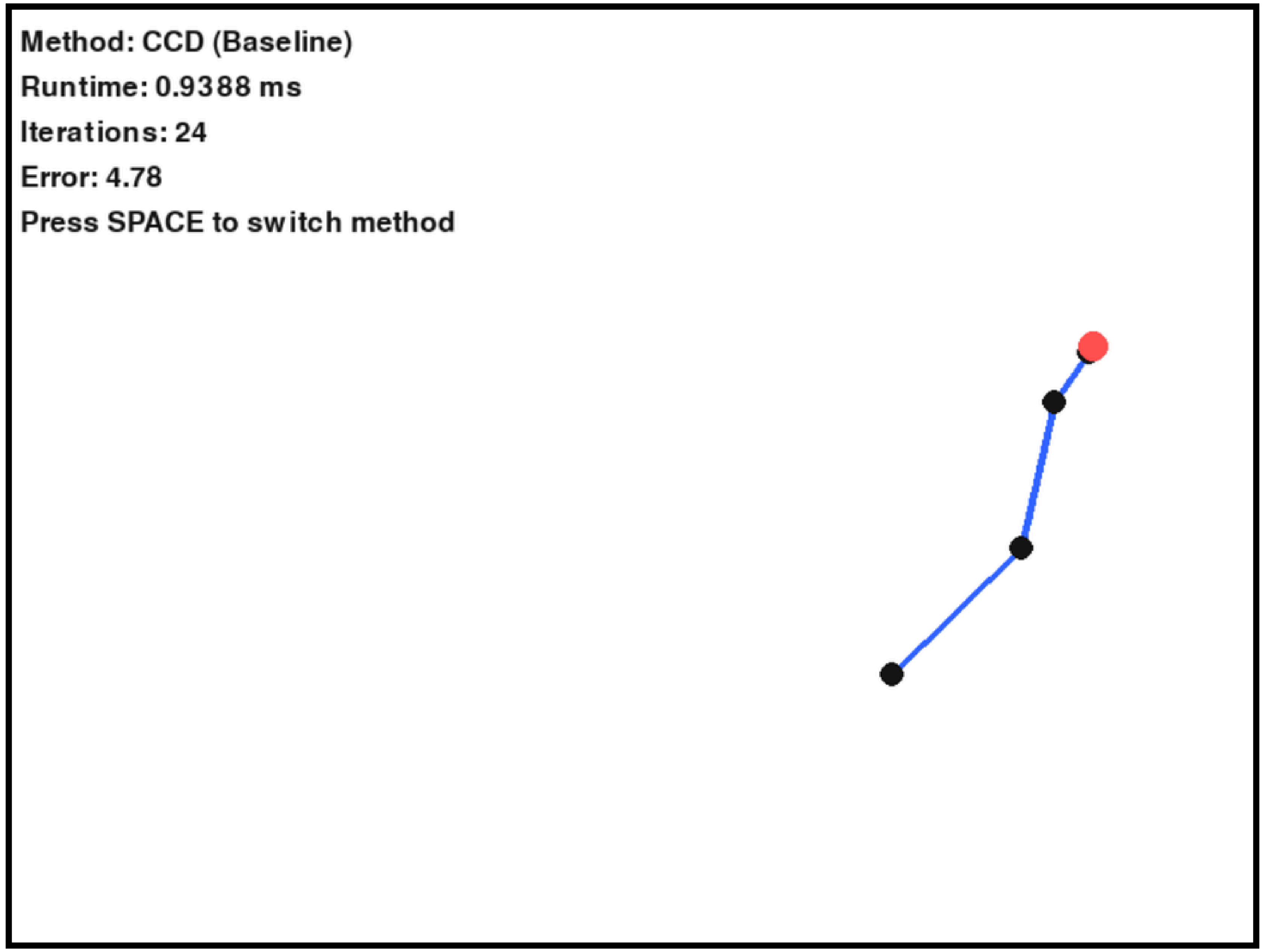
Real-Time Inverse Kinematics: CCD vs FABRIK

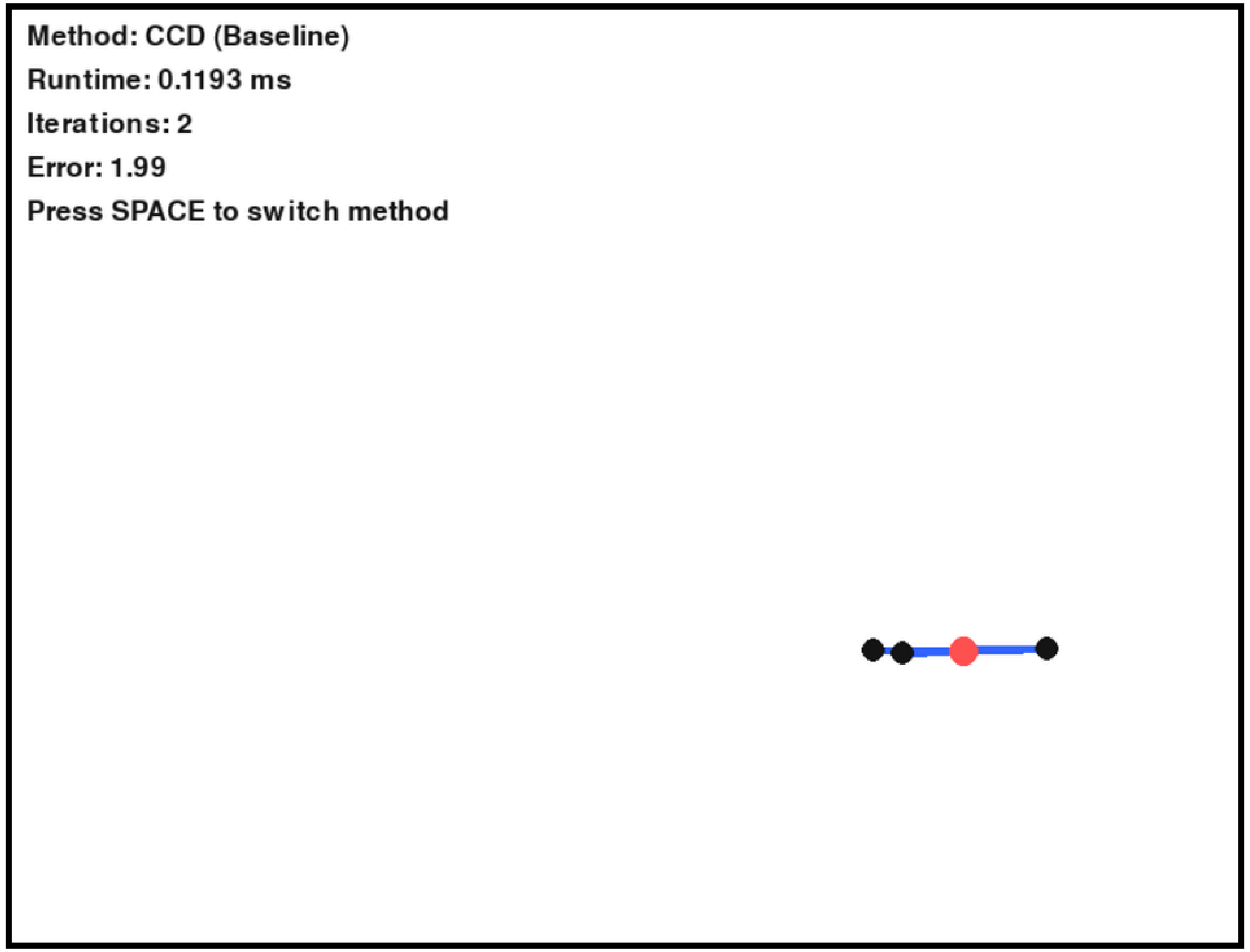




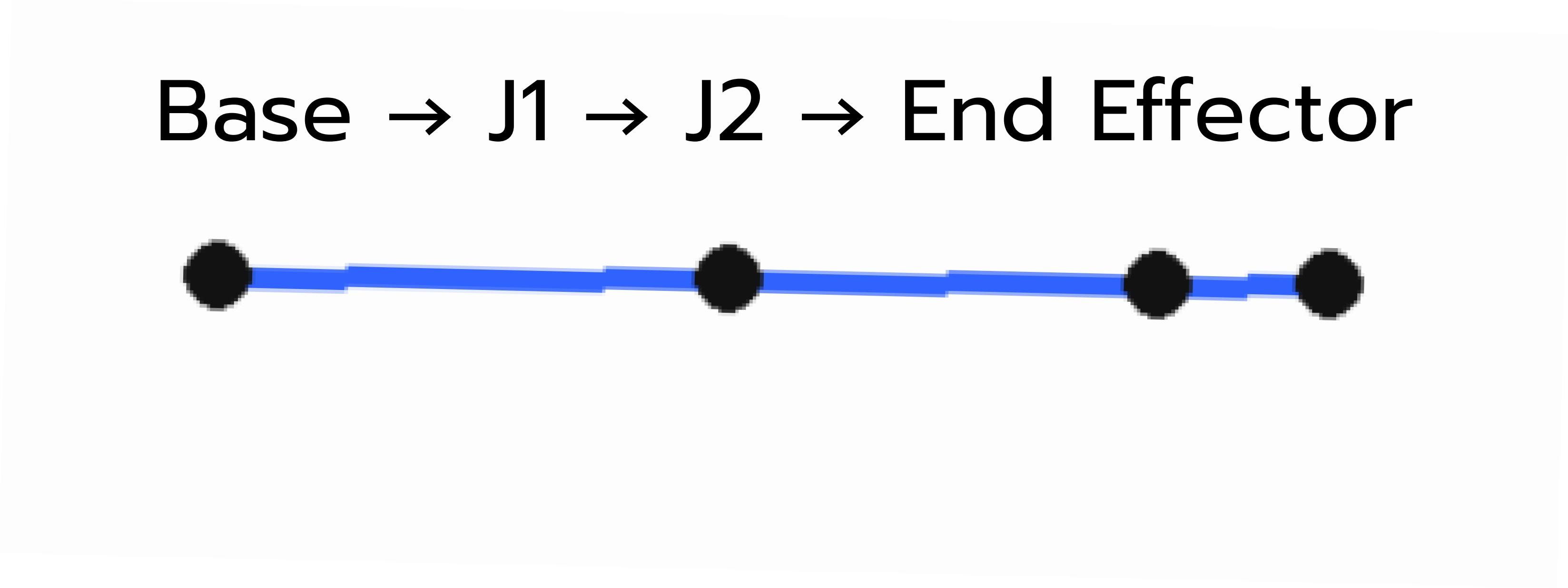
Problem Statement

- Requires multiple iterations to converge
- Can cause self-overlapping / arm folding
- Performance degrades with fast-moving target

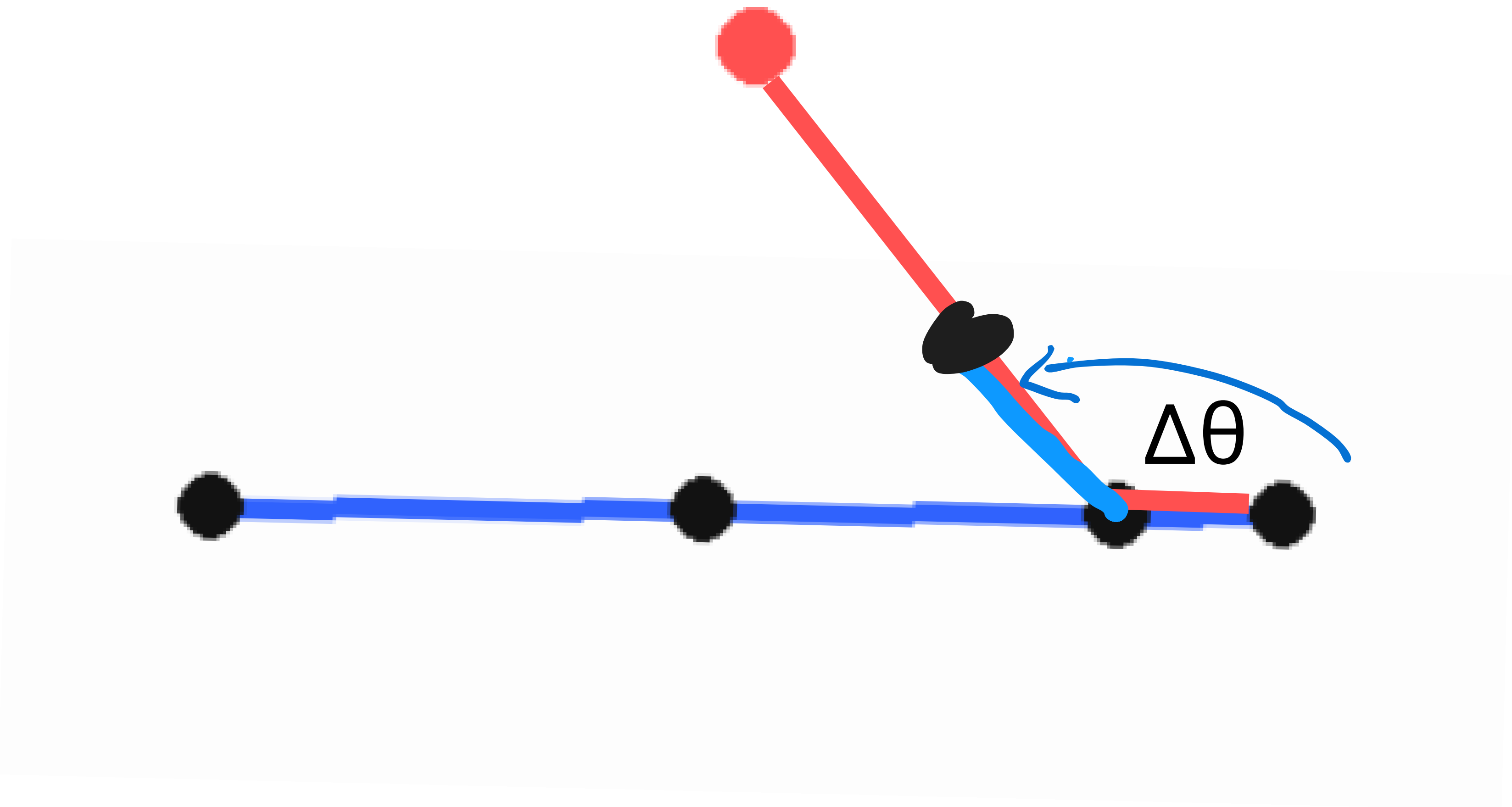




CCD



- Rotate one joint at a time
- Start from end joint
- Iterative angle update

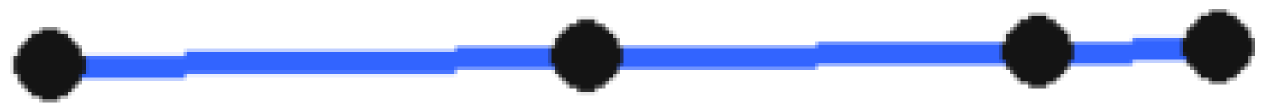


θ1 = มุมจาก joint → end effector
θ2 = มุมจาก joint → target

มุมที่ต้องหมุน = Δθ

FABRIK

Base → J1 → J2 → End Effector



Backward Pass

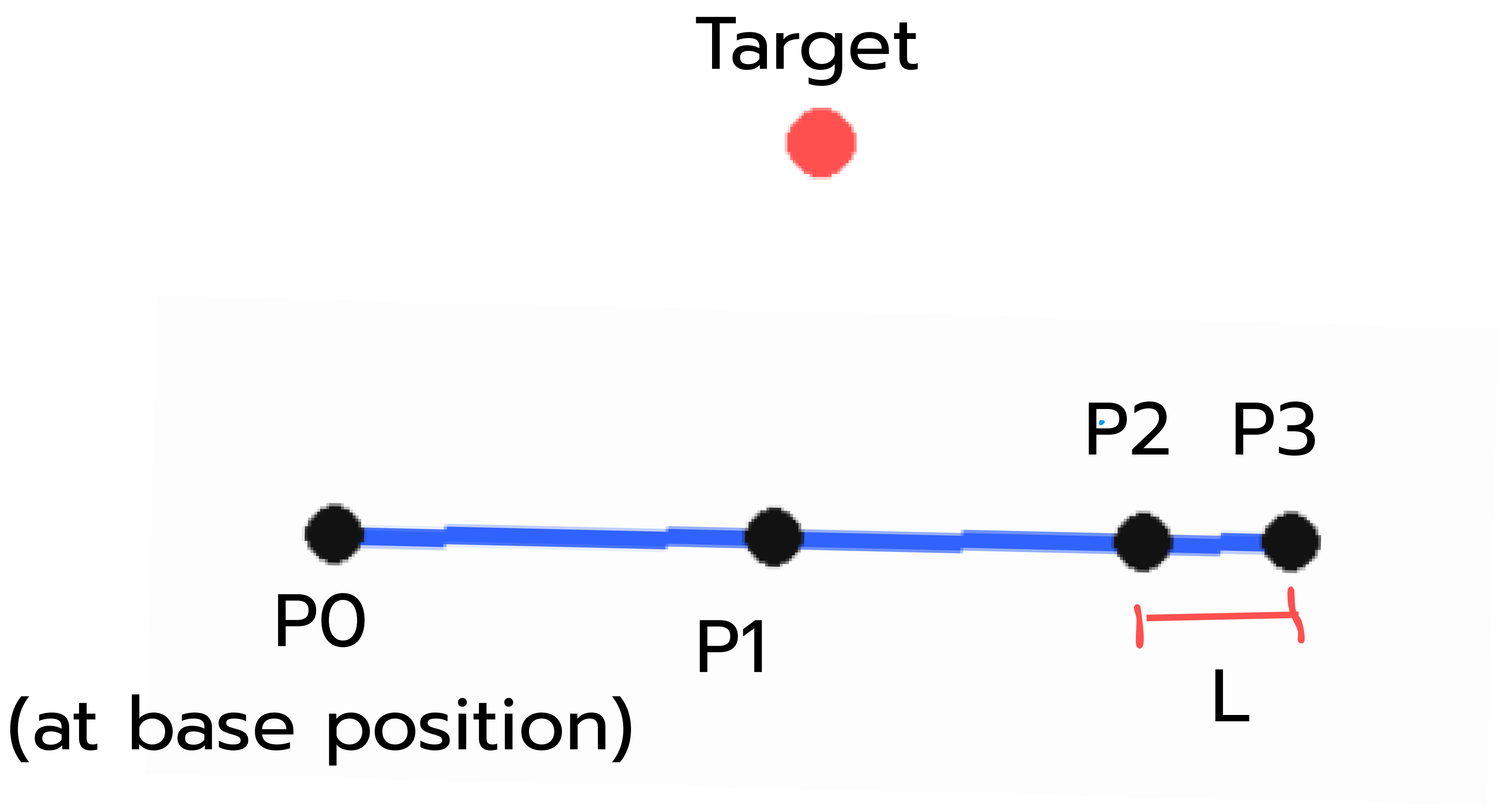
- End → Base

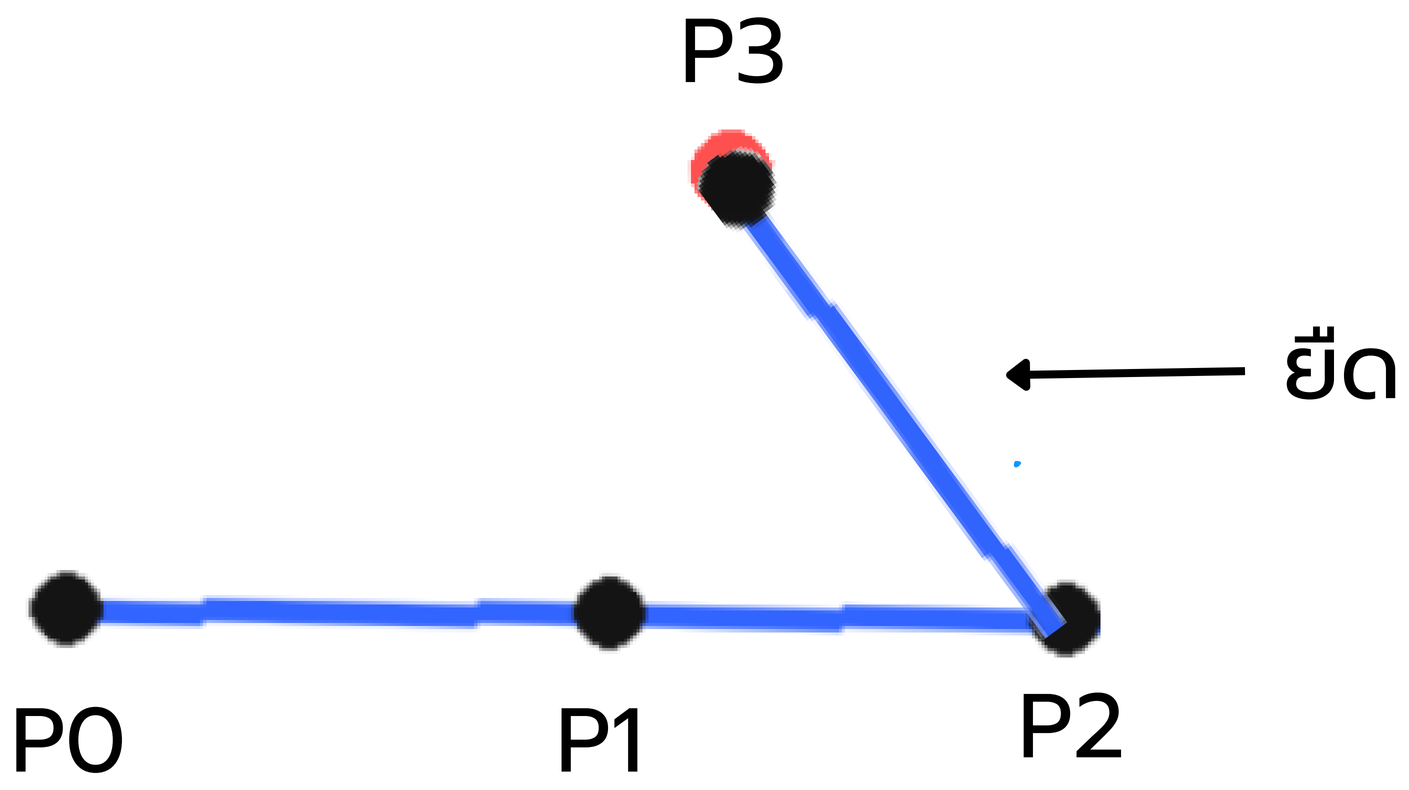
Forward Pass

- Base → End

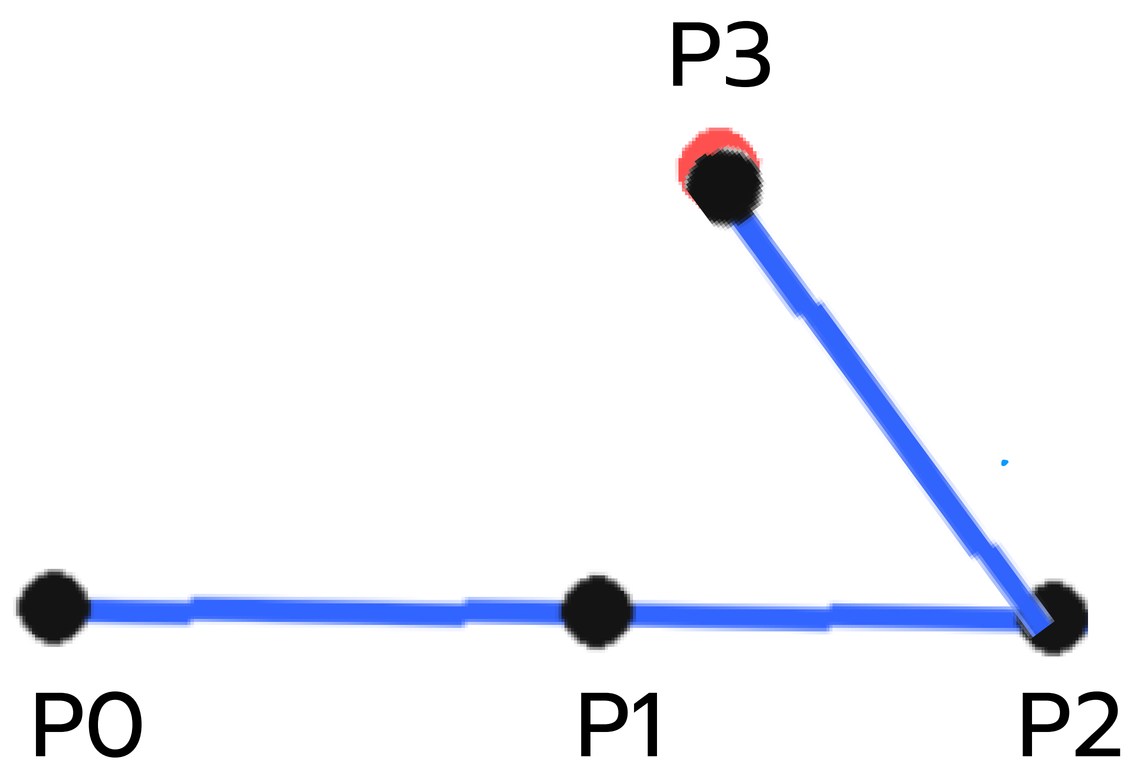
- Position-based method
- No direct angle rotation
- Global chain adjustment

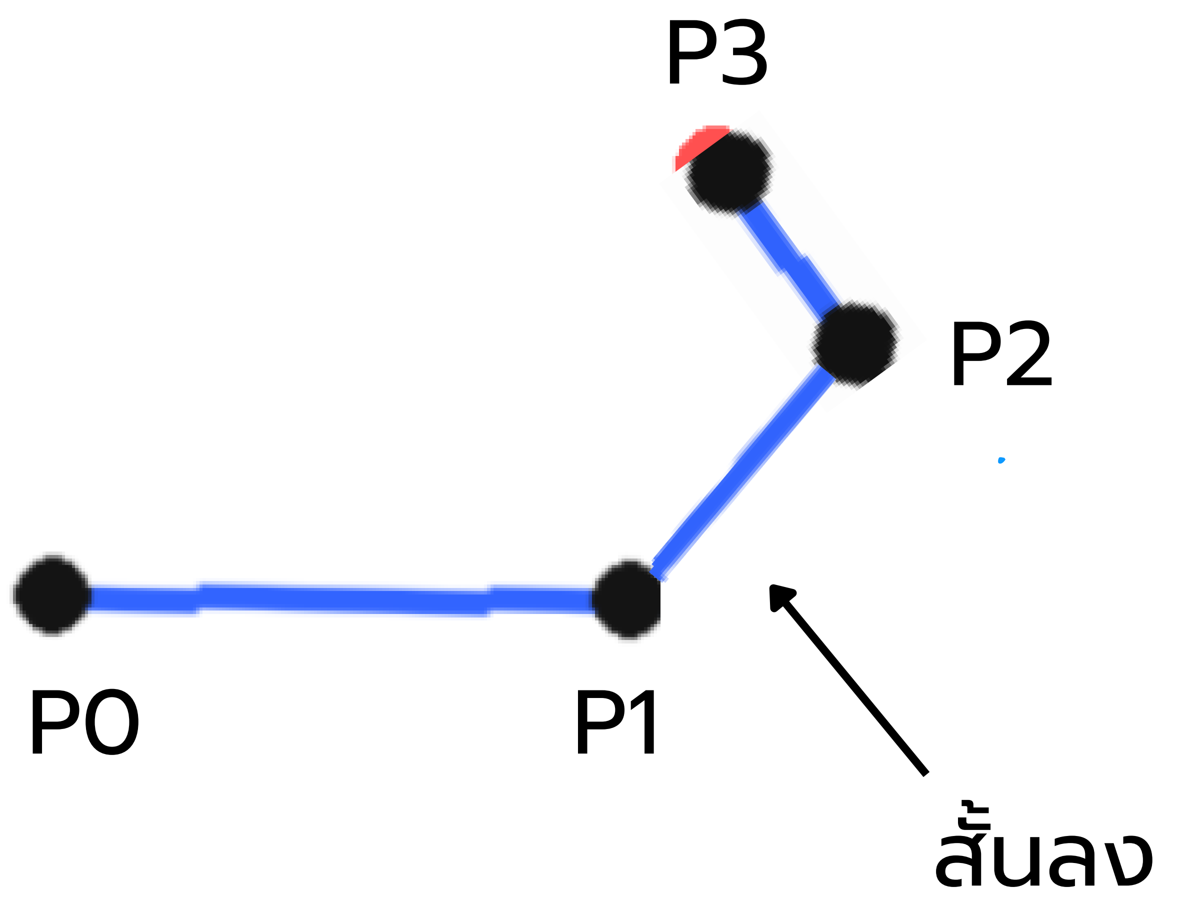
Backward Pass





P3
ยืด
P0
P1
P2

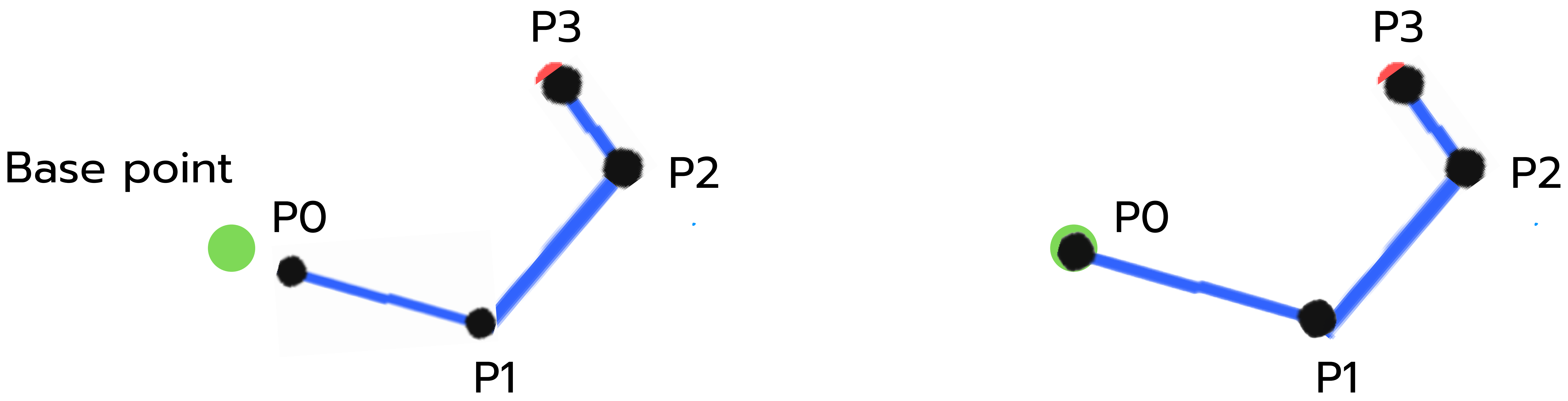




P2(new) = normalize(P3 - P2) * L

Pi(new) = normalize(Pi+1 - Pi) * Base Length(Pi, Pi+1)

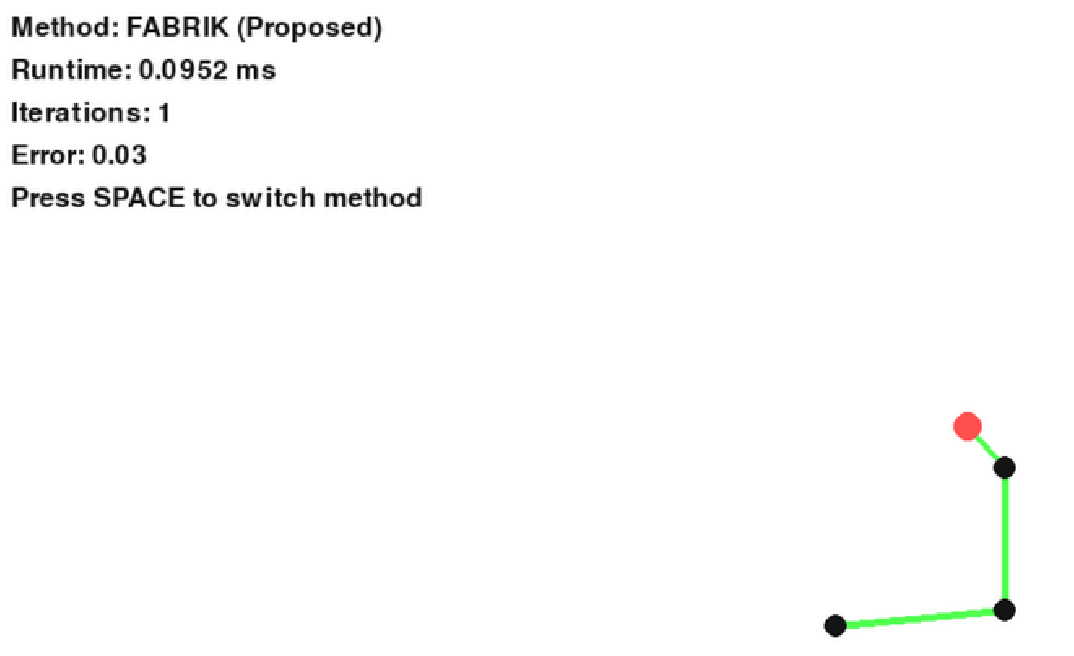
Forward Pass

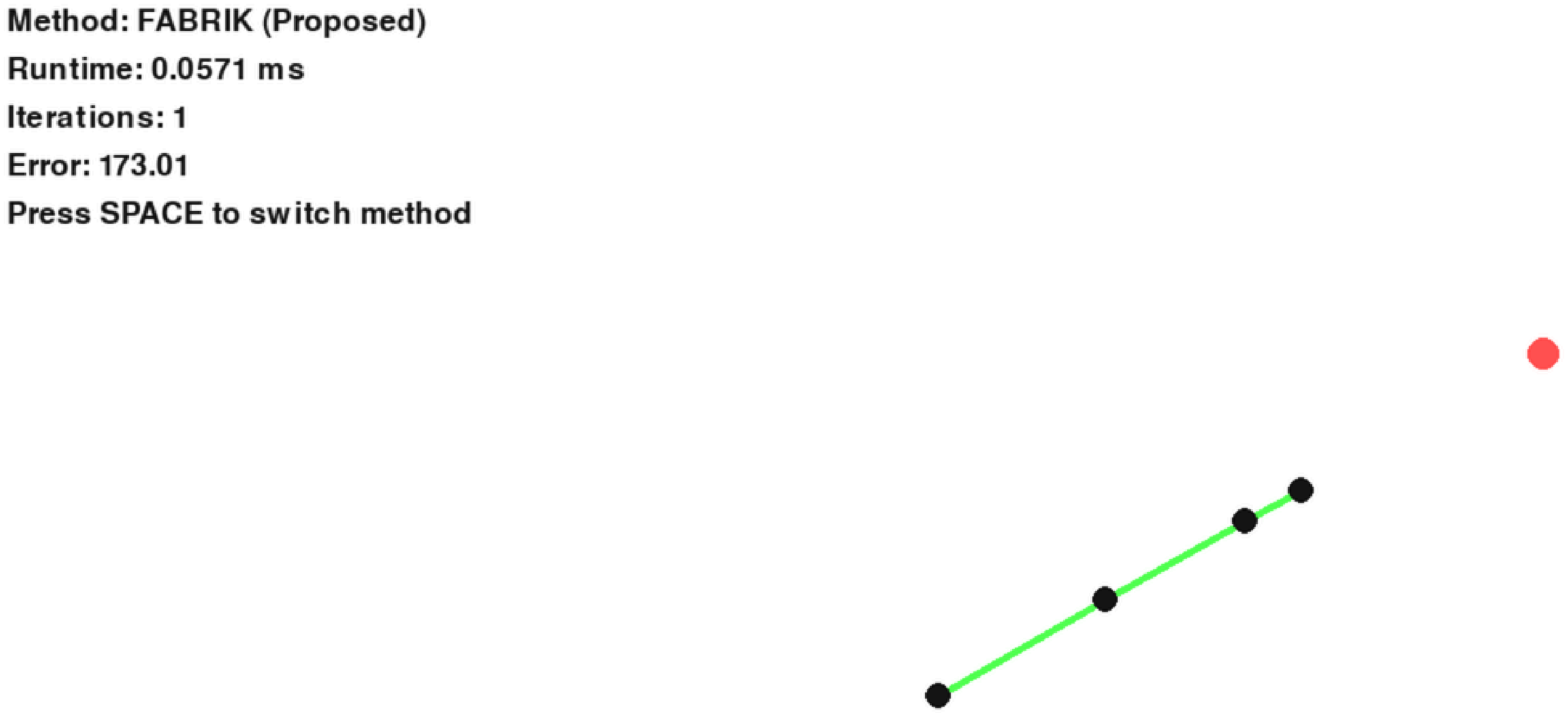


ทำเหมือนกับ Backward Pass แต่กลับด้าน

Early Stopping

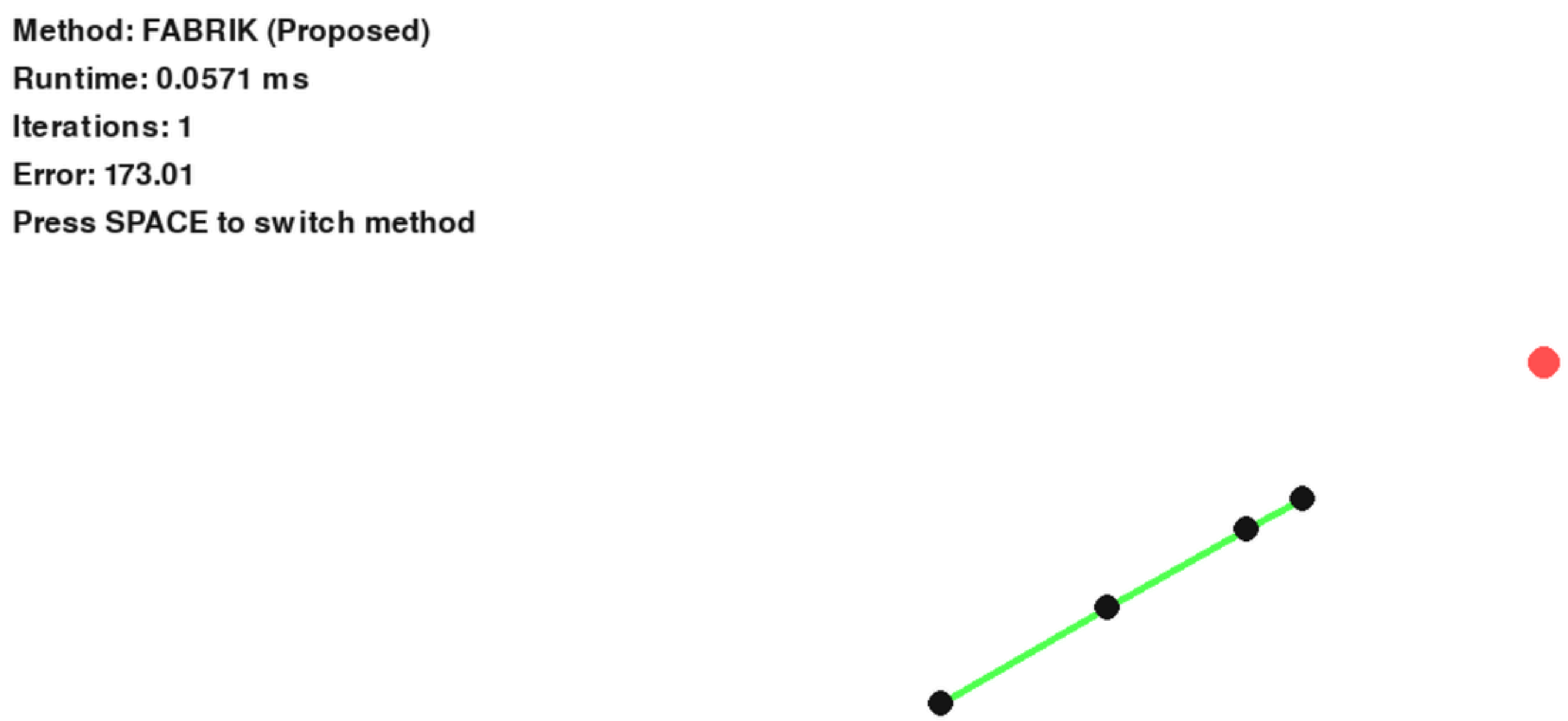
ถ้า error < threshold → หยุดการคำนวณ





Unreachable Target Clamping

ถ้า Target อยู่ห่างเกิน → ยืดแขนตรงทันที ไม่ต้องคำนวณเพิ่ม



Complexity

CCD

Per iteration:

- Loop through all joints → $O(n)$
- Recompute forward kinematics inside loop → $O(n)$

Total: $O(kn^2)$

FABRIK

Per iteration:

- · Backward pass → $O(n)$
- · Forward pass → $O(n)$

Total: $O(kn)$

Experiment Design

Environment

- Python 3 + Pygame
- 2D articulated arm
- 3, 10 segments
- Fixed base position
- Mouse-controlled target

Test Cases

- Case 1: Reachable target
- Case 2: Target close to base
- Case 3: Unreachable target

Result

Test Case	Method	Iterations	Runtime (ms)
Reachable Target	CCD	1-4	0.05–0.10
Reachable Target	FABRIK	1-2	0.04–0.05
Target Near Base	CCD	1-4	0.05–0.10
Target Near Base	FABRIK	1-2	0.04–0.05
Unreachable Target	CCD	30	Highest
Unreachable Target	FABRIK	1	Lowest

Result

Test Case	Method	Iterations	Runtime (ms)
Reachable Target	CCD	2-5	0.1-0.2
Reachable Target	FABRIK	1-2	0.07–0.09
Target Near Base	CCD	2-3	~0.1
Target Near Base	FABRIK	1-2	0.07–0.09
Unreachable Target	CCD	30	Highest
Unreachable Target	FABRIK	1	Lowest

Limitations

- No joint constraints → may produce unrealistic poses
- Limited to 2D motion
- FABRIK may have pose ambiguity in some cases

When CCD Can Be Better

- Easier to apply joint angle constraints
- Direct control over joint rotations
- Comparable performance for short chains
- More predictable behavior

Future Works

- Joint Constraint
- 3D
- Physics-based

Conclusion

- FABRIK converges faster and uses fewer iterations
- Produces smoother and more natural motion
- CCD provides better control over joint angles
- Easier to apply constraints with CCD
- CCD is more predictable in some cases

"No universally best method"